

6(a)	pH = 7.5 ; temperature = 55 °C ; A 51–57 °C	2
6(b)	any three from: at, optimum / pH 7.5 allow ecf from Q6(a) allows, maximum / 100%, activity of enzyme / active site functions at maximum rate ; AW A best fit of substrate into active site / AW either side of optimum hydrogen / ionic, bonds, affected / disrupted / broken ; R if, peptide / disulfide bond also stated may affect charges on, R-groups / side chains ; may affect ability to, transfer electrons (in catalytic site) ; may affect <u>shape</u> of active site / less ability to form enzyme-substrate complexes / <i>idea of</i> active site less complementary to substrate ; I ESC <i>answers may be in terms of denatured e.g. no enzyme-substrate complexes form</i> partial denaturation / some enzyme molecules denatured / starts to denature ; A enzymes for enzyme molecules A ref. to denatured at, pH 11 / 2% activity <i>must be a standalone idea</i> for this mp ignore responses such as 'from pH 7.5 to pH 11 activity decreases because enzyme is denatured'	3

Question	Answer	Marks
6(c)	any one from: 1 easier for test to take place at room temperature ; ora 2 ALT may denature (at optimum temperature, for LDH / at 55°) ; 3 substances reacting with ALT / substrates produced for LDH, may be affected ; 4 reaction may proceed too quickly to obtain accurate results / AW ; 5 bacterial LDH may be a different form ; 6 AVP ; e.g. suggestion that, reaction using ALT cannot occur / ALT may have different optimum temperature	1